BADS 6004
Managing Big data
Data Ingestion Lab

Importing from MySQL to HDFS

- Let first create a simple database under MySQL.

  # mysql –uroot –pcloudera

  mysql > create database energydata;

  mysql > use energydata;

  mysql > create table avgprice_by_state (

  year INT NOT NULL,

Importing from MySQL to HDFS

```
    state VARCHAR(5) NOT NULL,
    sector  VARCHAR(255),
    residential  DECIMAL(10,2),
    industrial   DECIMAL(10,2),
    transportation      DECIMAL(10,2),
    other          DECIMAL(10, 2),
    total          DECIMAL(10,2) );
mysql > quit;
```

Importing from MySQL to HDFS

cd /mnt
wget https://github.com/bbengfort/hadoop-fundamentals/raw/master/data/avgprice_kwh_state.zip
unzip avgprice_kwh_state.zip

Importing from MySQL to HDFS

```
# mysql -h localhost -uroot -pcloudera
  --local-infile=1
  mysql > use energydata;
  mysql > load data local infile
  '/mnt/avgprice_kwh_state.csv' into table
  avgprice_by_state fields terminated by ',' lines
  terminated by '\n' ignore 1 lines;
  mysql > quit;
```

Importing from MySQL to HDFS

```
# sqoop import --connect
    jdbc:mysql://localhost:3306/energydata
    --username root  --password cloudera --table
    avgprice_by_state --target-dir
    /home/cloudera/energydata  -m  1

# hadoop fs -cat /home/cloudera/energydata/part-
    m-00000
```

Importing from MySQL to Hive

sqoop import --connect jdbc:mysql://localhost:3306/energydata --username root --password cloudera --table avgprice_by_state --hive-table avgprice --hive-import -m 1

hive

hive > select * from avgprice;

hive > select * from avgprice where state=='CA';

Ingesting Product Impression Data

- Use Flume to consume the streaming user-interaction data generated by a hypothetical online store.
- Simulate an ecommerce impression log that records user interactions in the following JSON format:


```
{
    "sku" : "T9921-5"
   "timestamp" : 1453167527737
   "cid" : "51761"
   "action" : "add_cart"
   "ip" : "226.43.51.25"
}
```

Ingesting Product Impression Data

- To create the necessary directories and HDFS, download and run the script as a user with sudo privileges

  # cd /mnt

  # wget https://raw.githubusercontent.com/bbengfort/hadoop-fundamentals/master/flume/setup.sh)

Ingesting Product Impression Data

- Use nano editor to edit the setup.sh as follow :

```
#!/bin/bash
hadoop fs -mkdir -p /user/cloudera/impressions/
hadoop fs -chmod 1777 /user/cloudera/impressions/

mkdir /tmp/impressions
chmod  777 /tmp/impressions
mkdir  /tmp/flume
chmod 1777 /tmp/flume
```

- Execute the setup file by

```
# sh setup.sh
```

Ingesting Product Impression Data

- Download and edit a python program to create the log file named impressions.log at /tmp/impressions

  # cd /mnt

  # wget https://raw.githubusercontent.com/bbengfort/hadoop-fundamentals/master/flume/impression_tracker.py

- Add execution privilege to the file then execute it as follow :

  # chmod +x impression_tracker.py

  # ./impression_tracker.py

Download Examples of Configuration files

- Download the configuration files, one for client agent and one for collector agent

  # cd /mnt

  # wget https://raw.githubusercontent.com/bbengfort/hadoop-fundamentals/master/flume/client.conf

  # wget https://raw.githubusercontent.com/bbengfort/hadoop-fundamentals/master/flume/collector.conf

Configure Source Agent

```
# define spooling directory source :
client.sources=r1
client.sources.r1.channels=ch1
client.sources.r1.type=spooldir
client.sources.r1.spoolDir=/tmp/impressions

# define a file channel:
client.channels=ch1
client.channels.ch1.type=FILE

# define an Avro sink:
client.sinks=k1
client.sinks.k1.type=avro
client.sinks.k1.hostname=localhost
clinet.sinks.k1.port=4141
client.sinks.k1.channel=ch1
```

Configure Source Agent

```
# define an Avro sink:
client.sinks=k1
client.sinks.k1.type=avro
client.sinks.k1.hostname=localhost
clinet.sinks.k1.port=4141
client.sinks.k1.channel=ch1
```

Configure Collector Agent

```
# define an Avro source:
collector.sources=r1
collector.sources.r1.type=avro
collector.sources.r1.bind=0.0.0.0
collector.sources.r1.port=4141
collector.sources.r1.channels=ch1
```

Configure Collector Agent

```
# define a file channel using multiple disks for reliability
collector.channels=ch1
collector.channels.ch1.type=FILE
collector.channels.ch1.checkpointDir=/tmp/flume/checkpoint
collector.channels.ch1.dataDir=/tmp/flume/data

# define HDFS sinks to persist events as text
collector.sinks=k1
collector.sinks.k1.type=hdfs
collector.sinks.k1.channel=ch1
```

Configure Collector Agent

HDFS sink configuration

collector.sinks.k1.hdfs.path=/home/cloudera/impressions

collector.sinks.k1.hdfs.filePrefix=impressions

collector.sinks.k1.hdfs.fileSuffix=.log

collector.sinks.k1.hdfs.fileType=DataStream

collector.sinks.k1.hdfs.writeFormat=text

collector.sinks.k1.hdfs.batchSize=1000

Running Flume

flume-ng agent --name collector --conf . --conf-file ./collector.conf &

Ctrl+C

flume-ng agent --name client --conf . --conf-file ./client.conf

After finish importing, check for the imported files in the target directory

hadoop fs -ls /home/cloudera/impressions

Use
hadoop fs -cat /home/cloudera/impressions/impressions...log to display one of the imported files.

E.g., hadoop fs -cat /home/cloudera/impressions/impressions.1658357816743.log